

Exploring eBPF and its Integration with Kubernetes

eBPF, a game-changing technology that extends the capabilities of the Linux kernel, offers significant advantages for Kubernetes networking. It also greatly improves Kubernetes observability by capturing detailed telemetry data directly from the kernel. Read on to find out how its integration with Kubernetes has immense benefits.



eBPF (extended Berkeley Packet Filter) is a groundbreaking technology initially designed to enhance network packet filtering in the Linux kernel. Traditionally, BPF (Berkeley Packet Filter) could only be attached to sockets, with its first use case being network packet filtering for tools like tcpdump. In this case, BPF programs would filter packets by attaching to a raw AF_PACKET socket and displaying the filtered data.

However, eBPF has significantly evolved beyond its original scope. It now supports attaching to various kernel objects, extending its applicability far beyond socket filtering. Today, eBPF can hook into Kprobes, tracepoints, network schedulers (qdiscs) for classification and actions, and XDP (eXpress Data Path) for faster packet processing. These additions, along with newer features like in-kernel helper

functions and shared data structures (maps) that allow communication with user space, have transformed eBPF into a powerful tool for networking, observability, and security.

With these enhanced capabilities, eBPF offers precise control over system behaviour and observability in real time, making it an ideal technology to integrate with Kubernetes, which manages complex, distributed cloud-native environments.